Akoizo Original Practice - Form 1 - Answer key

Division C | Botany | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) NPP equals…

Unit: Botany, evidence, and ecosystem importance.
Worked-case data: A plant community fixes 100 carbon units and its producers respire 40 units over the same interval. Calculate NPP.

A. Gross production minus producer respiration

B. Gross production plus all respiration

C. Only nighttime growth

D. Total ecosystem carbon storage

Correct: A

2. (2 points) Higher water potential: -0.4 or -0.8 MPa?

A. -0.4 MPa

B. -0.8 MPa

C. They are identical

D. Neither can be compared

Correct: A

3. (2 points) Released photosynthetic oxygen originates from…

Unit: Photosynthesis, carbon fixation, and plant biochemistry.
Worked-case data: A leaf fixes 12 relative carbon units and releases 4 through the specified respiratory balance. What is net gain?

A. Water

B. CO_2 directly

C. Glucose only

D. Nitrogen gas

Correct: A

4. (2 points) Hydroponic roots still need…

Unit: Horticulture, aquatic systems, medicine, and materials.
Worked-case data: Two irrigation treatments produce the same growth index, but one uses twice the water. What else is needed to choose a management approach?

A. Oxygen and suitable nutrient conditions

B. No oxygen

C. Only pure water forever

D. No selective membranes

Correct: A

5. (2 points) Guard cells regulate…

Unit: Roots, stems, and leaves.
Worked-case data: A leaf closes its stomata during dry conditions. Predict one benefit and one cost.

A. Root hair length only

B. Stomatal opening

C. Every xylem vessel directly

D. Pollen chromosome number

Correct: B

6. (2 points) The example most strongly supports…

Unit: Major plant groups and comparative identification.
Worked-case data: A specimen has parallel leaf venation, floral parts in threes, and scattered stem bundles. Identify the supported broad group.

A. Every gymnosperm

B. Monocot

C. Moss

D. Fern

Correct: B

7. (2 points) Do all plants directly fix atmospheric nitrogen?

Unit: Plant ecology, nutrient deficiencies, and disease.
Worked-case data: Older leaves yellow first, but soil is dry and no tissue analysis exists. Is nutrient deficiency established?

A. Only every woody stem

B. No

C. Yes

D. Only every green leaf

Correct: B

8. (2 points) Phloem flow is best described as…

Unit: Plant cells, tissues, and vascular organization.
Worked-case data: A mature leaf exports sugar to a growing root. Identify source, sink, and transport tissue.

A. Always upward

B. Only from roots to leaves

C. Source to sink

D. Always downward

Correct: C

9. (2 points) Expected recessive phenotype in Aa x Aa with complete dominance?

Unit: Life cycles, flowers, seeds, and biotechnology.
Worked-case data: Track ploidy in the standard angiosperm pathway: diploid sporophyte, spore, gametophyte, gamete, zygote.

A. 3/4

B. All offspring

C. 1/4

D. 1/2

Correct: C

10. (2 points) Similar traits can arise independently through…

Unit: Plant evolution, fossils, and diversity.
Worked-case data: Two modern lineages share a vascular-tissue ancestor. Does one living lineage have to be the ancestor of the other?

A. A display stretch

B. A universal ladder

C. Convergence

D. Only identical ancestry

Correct: C

11. (4 points) Calculate the example NPP.

Unit: Botany, evidence, and ecosystem importance.
Worked-case data: A plant community fixes 100 carbon units and its producers respire 40 units over the same interval. Calculate NPP.

answer: 100 - 40 = 60 carbon units per stated interval. (4 points).

12. (4 points) Predict movement from -0.2 to -0.5 MPa.

answer: Toward -0.5 MPa along an available pathway, from higher to lower potential. (4 points).

13. (4 points) Calculate the example net gain.

Unit: Photosynthesis, carbon fixation, and plant biochemistry.
Worked-case data: A leaf fixes 12 relative carbon units and releases 4 through the specified respiratory balance. What is net gain?

answer: 12 - 4 = 8 relative units over the same interval. (4 points).

14. (4 points) Distinguish hydroponics and aquaculture.

Unit: Horticulture, aquatic systems, medicine, and materials.
Worked-case data: Two irrigation treatments produce the same growth index, but one uses twice the water. What else is needed to choose a management approach?

answer: Hydroponics is a plant cultivation method using nutrient solution; aquaculture cultivates aquatic organisms and can include different systems. (4 points).

15. (4 points) Explain the stomatal tradeoff.

Unit: Roots, stems, and leaves.
Worked-case data: A leaf closes its stomata during dry conditions. Predict one benefit and one cost.

answer: Closing pores conserves water but can restrict CO_2 entry and hence photosynthesis. (4 points).

16. (4 points) Why avoid one-trait identification?

Unit: Major plant groups and comparative identification.
Worked-case data: A specimen has parallel leaf venation, floral parts in threes, and scattered stem bundles. Identify the supported broad group.

answer: Traits have exceptions and can converge; combined reproductive and anatomical evidence is more reliable. (4 points).

17. (4 points) Why can older leaves show deficiency first?

Unit: Plant ecology, nutrient deficiencies, and disease.
Worked-case data: Older leaves yellow first, but soil is dry and no tissue analysis exists. Is nutrient deficiency established?

answer: Some mobile nutrients are redistributed to new growth, leaving older tissues depleted. (4 points).

18. (4 points) Why are lignified xylem walls useful?

Unit: Plant cells, tissues, and vascular organization.
Worked-case data: A mature leaf exports sugar to a growing root. Identify source, sink, and transport tissue.

answer: They support the plant and resist collapse under the tension of water transport. (4 points).